

Week 4

Monday 2/1

Explicit Runge Kutta (ERK)

$$w_{i+1} = w_i + h \sum_{j=1}^v b_j f(t_i + c_j h, z_j)$$

$$z_j = w_i + h \sum_{k=1}^{j-1} a_{jk} f(t_i + c_k h, z_k), \quad j=1, \dots, v-1$$

$j=1$:

$$\begin{aligned} w_{i+1} &= w_i + h \sum_{j=1}^v b_j f(t_i + c_j h, z_j) \\ &= w_i + h b_1 f(t_i, z_1) \end{aligned}$$

how to find unknowns (a_s, b_s, c_s)

→ Taylor expand things like $f(t+\Delta t, y+\Delta y)$

→ Since this is local, assume $w_i = y_i$

→ compare method to Taylor expansion of y_{i+1}

Let $v=1$

$$y_{i+1} = y_i + h y_i' + \frac{h^2}{2} y_i'' + \frac{h^3}{6} y_i''' + O(h^4)$$

$$\rightarrow y_{i+1} - w_{i+1} = h y_i' (1 - b_1) + O(h^2)$$

→ for this to be 0, $b_1 = 1$ (generally $1 = \sum_{j=1}^v b_j$)

y in HW2

table:

0	0
1	1

← no a variable

Euler's Method is 1st order ERK

$$\text{for ERK2} - f\left(t_i + \overbrace{c_2 h}^{\Delta t}, y_i + \overbrace{a_{21} h f(t_i, z_1)}^{\Delta y}\right)$$

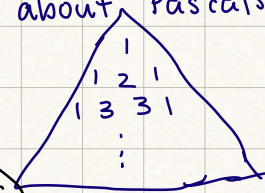
Taylor expand $f(t+\Delta t, y+\Delta y)$ about (t, y)

Think about Pascal's Triangle

$$= f(t, y) + (\Delta t f_t + \Delta y f_y) / 1!$$

$$+ \frac{1}{2!} (\Delta t^2 f_{tt} + 2\Delta t \Delta y f_{ty} + \Delta y^2 f_{yy})$$

$$+ \frac{1}{3!} (\Delta t^3 f_{ttt} + 3\Delta t^2 \Delta y f_{tty} + 3\Delta t \Delta y^2 f_{tyy} + \Delta y^3 f_{yyy}) \dots \text{etc}$$



Most popular is ERK4;

c_k		a_{jk}			
c_1	0	-			
c_2	$\frac{1}{2}$	$\frac{1}{2}$	-		
c_3	$\frac{1}{2}$	0	$\frac{1}{2}$	-	
c_4	1	0	0	1	-
b_j	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$	

$$\begin{aligned}
 w_{i+1} = & w_i + h \left(\frac{1}{6} f(t_i, y_i) + \frac{1}{3} f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2} z_1\right) \right. \\
 & + \frac{1}{3} f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2} z_2\right) \\
 & \left. + \frac{1}{6} f(t_i + h, y_i + h z_3) \right)
 \end{aligned}$$

$z_2 \approx y(t_i + \frac{h}{2}) = y_i + \frac{h}{2} z_1$
 $z_3 \approx y(t_i + h) = y_i + h z_2$

this method has a local error $O(h^5)$ (global is $O(h^4)$)

REVIEW 3 ★

Why is ERK4 so popular?

	RK2	RK3	RK4	RK n	RK n
stages/substeps	2	3	4	$5 \leq n \leq 7$	$8 \leq n \leq 9$
best possible order	$O(h^2)$	$O(h^3)$	$O(h^4)$	$O(h^{n-1})$	$O(h^{n-2})$

← # of func evaluations

Wednesday 2/4

(linear) multistep methods (LMM)

GOALS: $\square$ define LMM

$\square$ stencils

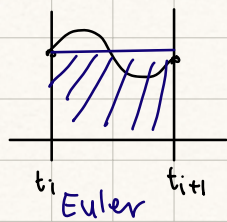
$\square$ derivation of methods

$\square$ root conditions

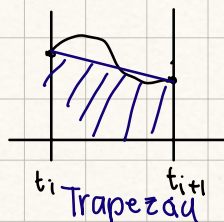
IVP: $\frac{dy}{dt} = f(t, y); y(t_0) = y_0$

Integrate: $\Rightarrow y(t_{i+1}) = y(t_i) + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt$

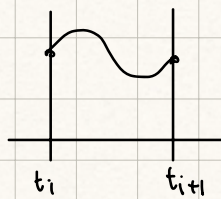
IDEA: Approx $f(t, y(t))$ using a Lagrange int. poly.



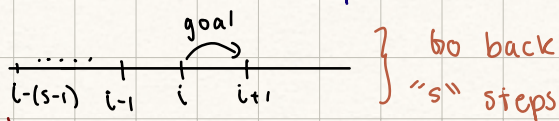
Euler



Trapezoidal



... what if we go beyond...



allows us to build Lagrange Int Polynomials assuming equally spaced nodes

$$p(t) = f(t, y(t)) + O(h^s)$$

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} p(t) dt + O(h^{s+1})$$

note: need another method to generate data at $t_i, t_{i-1}, \dots, t_{i-s-1}$

$$\int_{t_i}^{t_{i+1}} \sum_{k=0}^{s-1} L_k(t) f(t_{i-k}, y_{i-k}) dt$$

recall:

$$L_k(t) = \frac{(t-t_1)(t-t_2)\dots(t-t_k)}{(t_0-t_1)(t_0-t_2)\dots(t_0-t_k)}$$

* Note: need to review this $\hat{=}$

HOW THE BOOK DOES IT

$$\left\{ \begin{array}{l} b_k = \frac{1}{h} \int_{t_i}^{t_{i+1}} L_k(t) \end{array} \right.$$

→ b_k are INDEPENDENT of timestep i & choice of h → robustness

→ after some work:

$$w_{i+1} = w_i + b_0 f(t_i, w_i) + b_1 f(t_{i-1}, w_{i-1}) + \dots + b_{s-1} f(t_{i-(s-1)}, w_{i-(s-1)})$$

generalize this to the following:

$$w_{i+1} = a_0 w_i + a_1 w_{i-1} + \dots + a_{s-1} w_{i-(s-1)}$$

} y values

$$\underbrace{b_{-1} f(t_{i+1}, w_{i+1}) + b_0 f(t_i, w_i) + b_1 f(t_{i-1}, w_{i-1}) + \dots + b_{s-1} f(t_{i-(s-1)}, w_{i-(s-1)})}_{\text{general "s-step" method}} \} \text{derivative values}$$

s-step method stencils:

	y	y' = f	RESIDUAL FORM
i+1	1	-b ₋₁	
i	-a ₀	-b ₀	
i-1	-a ₁	-b ₁	
...	...	...	
i-(s-1)	-a _{s-1}	-b _{s-1}	

local analysis

$$y_{i+1} - w_{i+1} = "0"$$

for error analysis, throw

negatives in front

How to compute a_k & b_k ? TAYLOR EXPANSION ☺ $\approx$ about t_i

$$y_{i+1} = y_i + h y_i' + \frac{h^2}{2} y_i'' + O(h^3)$$

$$w_{i+1} = a_0 y_i +$$

$$+ a_1 (y_i - h y_i' + \frac{h^2}{2} y_i'' + O(h^3))$$

alt signs

$$+ b_{-1}$$

Friday 2/6

$$w_{i+1} = a_0 w_i + a_1 w_{i-1} + \dots + a_{s-1} w_{i-(s-1)}$$

} y values

$$h(b_{-1} f(t_{i+1}, w_{i+1}) + b_0 f(t_i, w_i) + b_1 f(t_{i-1}, w_{i-1}) + \dots + b_{s-1} f(t_{i-(s-1)}, w_{i-(s-1)}))$$

general "s-step" method

} derivative values

in "residual form" take all terms to LHS (negative)

→ How to determine values for a_k & b_k

↳ usual way: Taylor expand

→ 2-step method:

$$\begin{aligned} w_{i+1} &= a_0 y_i \\ &+ a_1 (y_i - h y_i' + \frac{h^2}{2} y_i'' + O(h^3)) \\ &+ b_{-1} (h y_i' + \frac{h^2}{2} y_i'' + O(h^3)) \quad \leftarrow \text{implicit term} \\ &+ b_0 (\dots) \end{aligned}$$

→ form $y_{i+1} - w_{i+1} = O(h^2)$

→ a better way: recall that the approx $y(t+h) = y(t) + h y'(t) + \dots + \frac{h^k}{k!} y^{(k)}(t)$ } is exact for a degree polynomial up to k

ex! a one-step explicit method

$$w_{i+1} = a_0 w_i + h b_0 f_i$$

residual form:

$$w_{i+1} - a_0 w_i - h b_0 f_i = 0$$

stencil:

set $h=1$ Always

	y	y'
i+1	1	0
i	-a ₀	-b ₀

usual tedious method

$$y_i + h y_i' + \frac{h^2}{2} y_i'' + O(h^3) - a_0 y_i - b_0 h y_i'$$

$$y_{i+1} - w_{i+1} = (1 - a_0) y_i + (1 - b_0) h y_i' + (1 - 0) \frac{h^2}{2} y_i'' + O(h^3)$$

→ $a_0 = 1$, $b_0 = 1$ then it is $O(h^2)$

instead, work w/ polynomials

	y	y'
$i+1$	1	0
i	$-a_0$	$-b_0$

$$W_{i+1} - a_0 W_i - b_0 f_i = 0$$

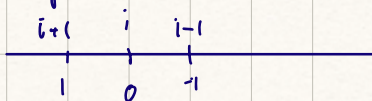
$\rightarrow$ test with monomials

	1	0
$i+1$	w_{i+1}	f_{i+1}
i	w_i	f_i

$$1 - a_0(1) = 0$$

$$a_0 = 1$$

note: our grid:



	x	1
$i+1$	1	1
i	0	1

$$1 - b_0 = 0$$

$$b_0 = 1$$

	x^2	$2x$
$i+1$	1	2
i	0	0

	y	y'
1	1	0
0	$-a_0 - b_0$	
-1	$-a_1 - b_1$	

	1	0
1	1	0
0	1	0
-1	1	0

	x	1
1	1	1
0	0	1
-1	-1	1